

VTP Protocol:

VTP (VLAN Trunking Protocol) is a Cisco proprietary protocol used in networking to manage VLAN configurations across multiple switches.

Purpose of VTP

VTP helps to manage the addition, deletion, and renaming of VLANs across a network with multiple switches, ensuring that VLAN information is consistent throughout the network. It reduces the administrative overhead in large networks by centralizing VLAN management.

VTP Domain and Modes

VTP Domain: A collection of switches under the same administrative control. All switches within a VTP domain share VLAN information.

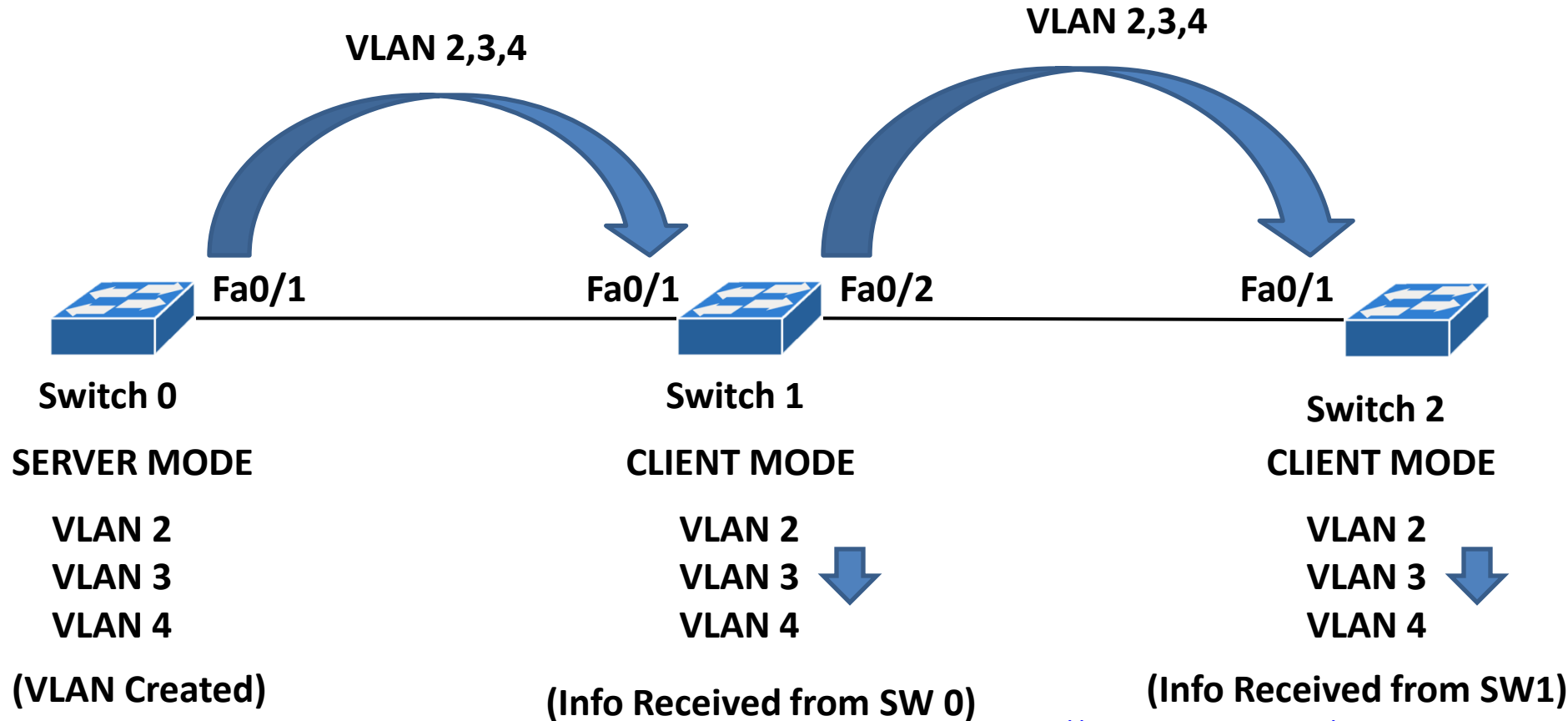
VTP Modes:

Server Mode: Switches in this mode can create, modify, delete, update and propagate VLAN changes to all other switches in the domain.

Client Mode: Switches in this mode cannot create, modify, or delete VLANs, but propagate VLAN changes to all other switches in the domain.

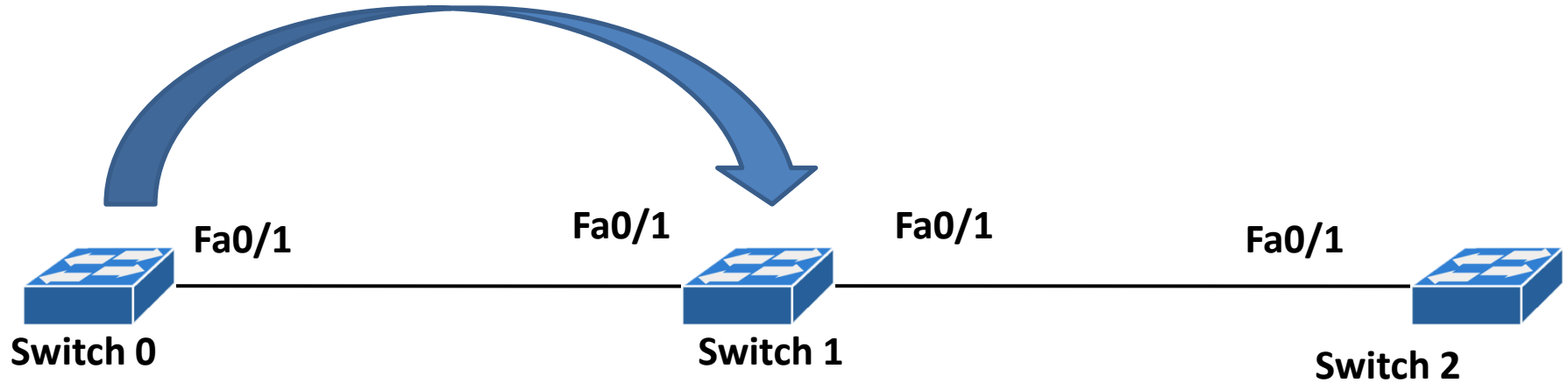
Transparent Mode: They can create, modify, and delete VLANs locally but do not propagate this information to other switches. They do forward VTP advertisements through their trunk links.

VTP Protocol: Configure VLAN Dynamically



SERVER MODE OF VTP

Propagate VLAN changes to all other switches



SERVER MODE

Create VLAN

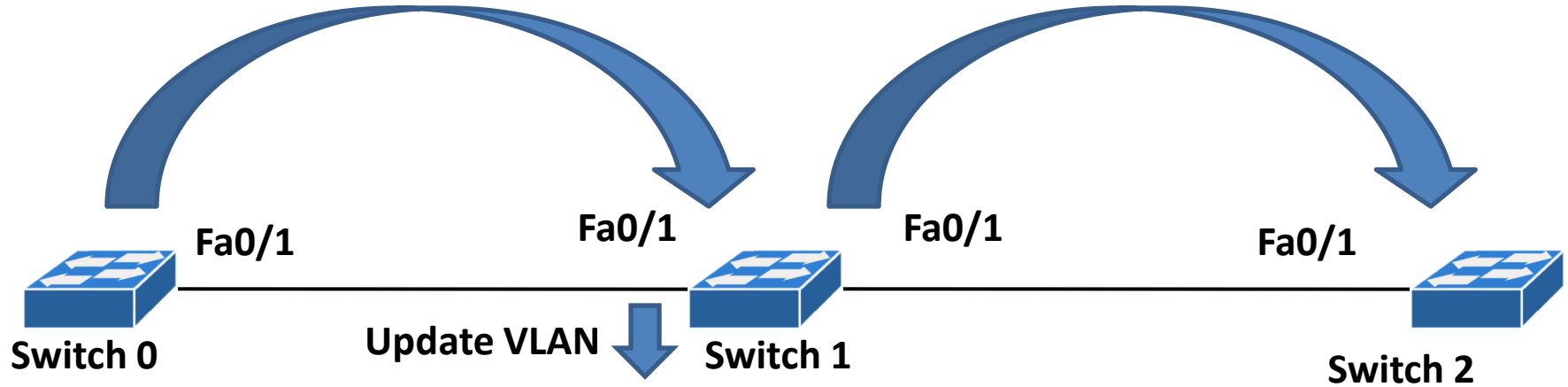
Delete VLAN

Update VLAN

Propagate VLAN

CLIENT MODE OF VTP

Propagate VLAN changes to all other switches



CLIENT MODE

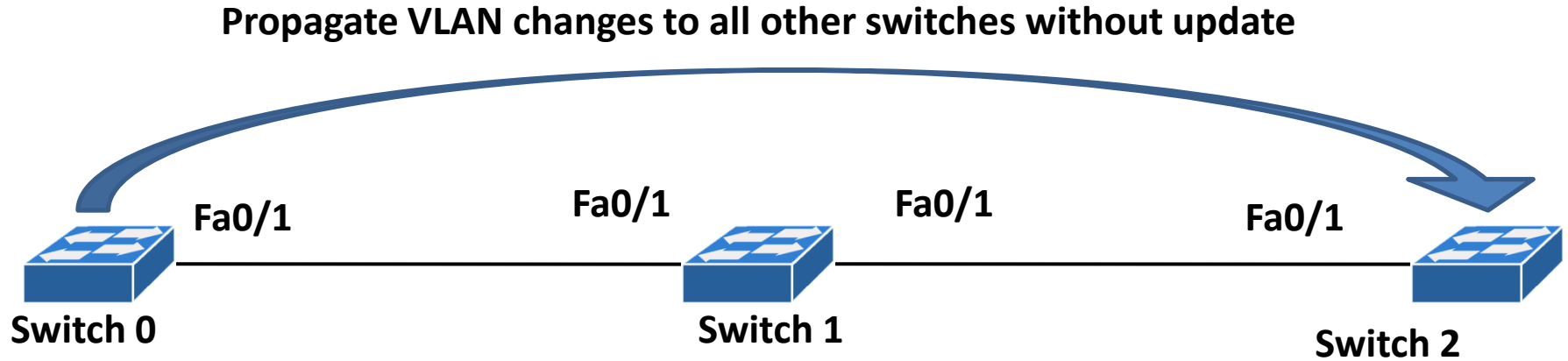
Update VLAN

Cant Create VLAN

Cant Delete VLAN

Propagate VLAN

Transparent MODE OF VTP



TRANSPARENT MODE

- Create VLAN Locally
- Delete VLAN Locally
- Cant Update VLAN
- Propagate VLAN

Comparison between VTP Modes

Server Mode	Client Mode	Transparent mode
Can Create VLAN	Can't Create VLAN	Can Create VLAN Locally
Can Delete VLAN	Can't Delete VLAN	Can Delete VLAN Locally
Can Propagate VLAN for others to change	Can Propagate VLAN for others to change	Can Propagate VLAN for others to change without updating itself
Can't Access Extended VLAN	Can't Access Extended VLAN	Can Access Extended VLAN
Can Update VLAN if received	Can Update VLAN if received	Can't Update VLAN if received

VTP Configuration and Rules:

- All the links must be trunk links
- Domain name must match on all devices
- Password must match

Switch>enable

Switch#configure terminal

Switch(config)#vtp domain CorporateDomain

Switch(config)#vtp mode Server/Client/Transparent

Switch(config)#vtp version 2

Switch(config)#vtp password SecurePass123

Switch(config)#end

Switch#show vtp status